

A detailed synthetic biopsy and linked EGFR report, processed by unchanged local methods.

Original synthetic report

Patient name: Lena Varnell
Medical record number: 48271639
Date of birth: 06/18/1965
Ordering provider: Mira Norvale, MD
MOLECULAR PATHOLOGY REPORT
Accession: MP32-006381
Parent accession: SP32-041728
Collected: 09/10/2032 09:25
Received: 09/13/2032 11:05
Reported: 09/18/2032 14:20

SPECIMEN INFORMATION
Site: Right upper lobe lung
Diagnosis: Pulmonary adenocarcinoma
Material: Formalin-fixed paraffin-embedded core biopsy
Specimen part: A | Block: A1
Tumor cellularity: 40% | Macrodissection: Performed

MOLECULAR RESULTS
EGFR p.L858R detected; variant allele fraction 28%.
EGFR p.T790M detected; variant allele fraction 12%.

VARIANT DETAILS

Gene	Transcript	Exon	Coding change	Protein change
EGFR	NM_005228.5	21	c.2573T>G	p.L858R
EGFR	NM_005228.5	20	c.2369C>T	p.T790M

INTERPRETATION

Two EGFR substitutions are detected. Correlate with prior molecular testing and treatment history. The phase relationship of the two variants is not determined. Tumor-only testing does not establish whether a variant is somatic or germline.

QUALITY METRICS

DNA input: 42 ng | Mean target depth: 1540x
L858R locus depth: 1640x | T790M locus depth: 1240x
Target bases at >=250x coverage: 99.4%
All listed quality metrics are simulated for this illustration.

METHOD

Illustrative targeted DNA sequencing of EGFR exons 18–21.
Single-nucleotide variants and small insertions/deletions assessed.
Illustrative reporting threshold: 5% variant allele fraction.
No RNA fusion or copy-number assessment is included.

LIMITATIONS

Low tumor content, sampling, and variants below the reporting threshold may limit detection. A negative result would not exclude alterations outside the interrogated regions. No paired normal sample was analyzed. Assay specifications are illustrative only.

Electronically signed: Arlen Voss, MD
Lena Varnell | 48271639 | MP32-006381 | Page 1 of 1

Original synthetic report

Synthetic illustration, outside both benchmark cohorts. Actual local outputs; no hosted LLM was run. Repeated synthetic banners and page margins are cropped for display.

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Institution-developed pipeline

Patient name: **** *
***** number: *****
Date of birth: **/**/****
Ordering provider: **** *, MD
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Electronically signed: ***** *, MD
***** | ***** | ****-***** | Page 1 of 1
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Institution-developed pipeline

un. Repeated synthetic banners and page margins are cropped for display.

Refined local rules

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Date of birth: *****
Ordering provider: *****
*****
Accession: *****
Parent accession: *****
*****
*****
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Electronically signed: ***** MD
***** | ***** | ***** | Page 1 of 1

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Refined local rules